(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JUNE 2024

Paper Code: BS-104 **Subject: Applied Chemistry**

Time: 3 Hours **Maximum Marks: 60**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Do **any six** parts: **(2x6=12)**

(a) All coking coals are caking coals but all caking coals are not coking coals.

(b) Hydrocarbons that are poor gasoline fuels are quite good diesel fuels. Explain

(c) A eutectic mixture has a definete composition and asharpmelting point, yet it is not a compound. Justify.

(d) Describe the method of preparation of the polymer, Nylon-6, 6.

(e) Why is phosphate conditioning better than the carbon aye conditioning?

(f) A water sample contains 248 mg $CaSO_4$ per liter. Calculate the hardness in terms of $CaCO_3$ equivalent.

(g) Why impure metals corrode faster than pure metal under identical conditions?

(h) A pure metal rod half immersed in water starts corroding at the bottom. Give reason.

UNIT-I

Q2 (a) A sample of coal was tested for its calorific value using Bomb's Calorimeter. Following data was obtained.
Weight of coal burnt = 0.920 gm.
Weight of water taken = 550 gm.
Water equivalent of bomb and calorimeter = 2200 gm.
Rise in temperature = 2.42 °C
Fuse wire correction = 10 Cal
Acid Correction = 50 Cal
Assuming coal to contain C = 93%, H = 6% and ash = 1%, Calculate gross and net calorific value of coal if the latent heat of condensation of steam is 580 Cal/gm. **(4)**

(b) What are the advantages of catalytic cracking over thermal cracking? **(4)**

(c) What do you understand by the term knocking in IC engine? Explain the relation between chemical structure and knocking in petrol engine. **(4)**

Q3 (a) A sample of coal was analyzed as follows. Exactly 2.5 g was weighed into a silica crucible. After heating for an hour at 110 °C, the residue weighed 2.415 g. The crucible was then covered with a vented lid and strongly heated for exactly 7 min at 950 ± 20 °C. The residue weighed 1.528 g. The crucible was then heated without the cover, until a constant weight was obtained. The last residue was found to weigh 0.245 gm. Calculate the percentage of moisture, volatile matter, ash and fixed carbon in the coal sample. **(4)**

(b) What are gaseous fuels? Give some examples. What are the merits and demerits of gaseous fuels over solid and liquid fuels? **(4)**

P.T.O.

(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B. tech Second Semester, April, 2024

Paper Code: BS-104 Subject: Applied Chemistry

Time: 1 ½ Hrs. Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	A sample of coal c[illegible]tains C=63%, O=30%, H=6%, S=0.5%, N=0.2% and Ash=0.3%. Calculate the GCV of coal by Dulong's formula.	2	C[illegible] 1
(b)	What is Sweetening of Petrol (with reaction)?	2	CO-1
(c)	With the help of Phase Rule, find the Degree of Freedom of the following systems: (i) Ice & water in equilibrium (ii) Ag & Pb at the Eutectic point (iii) Saturated solution of NaCl (iv) Solid Ag & Soln. of Ag & Pb.	2	CO-2
(d)	Classify the following into Plastics, Elastomers & Fibres: (i)Nylon-6 (ii) HDPE (iii) Natural Rubber (iv) PVC	2	C0-2
(e)	Giv Composition, properties (any 2), Uses (any 2) of Water Gas & Producer Gas.	2	CO-1
Q 2 (a)	Describe how the GCV & NCV of a Solid fuel is determined using a Bomb Calorimeter (with diagram). Also give the formula of GCV with corrections involved.	5	CO-1
(b)	A gaseous fuel has the following composition by volume – CH_4=6%, CO=24%, CO_2=10%, H_2=10%, N_2=50%. If 30% excess air is supplied, calculate the volume of air supplied for 1m^3 of the fuel.	3	CO-1
(c)	Differentiate between Thermal Cracking & Catalytic Cracking.	2	CO-1
Q 3 (a)	With the help of a well labelled diagram explain the Sulphur System. Also explain why in a Sulphur System all the four phases cannot exist in equilibrium.	5	CO-2
(b)	Explain about Eutectic System, Eutectic mixture & Eutectic point.	3	CO-2
(c)	Write about the following polymers (i) Conducting polymers (II) Biodegradable Polymers (iii) LDPE (on any 2).	2	C0-2
Q 4.	Attempt any 2 parts		
(a)	What is the reason for Knocking in IC Engine. Define Octane No. & Cetane No.	5	CO-1
(b)	2.12g of the coal sample was Kjehldahlised and NH_3 gas thus evolved was absorbed in 30ml of 0.1N H_2SO_4. After absorption the excess acid required 10ml of 0.1N NaOH for complete neutralisation. Determine the percentage of Nitrogen.	5	CO-1
(c)	Explain about the Binary System with Congruent melting point & how it is different from Eutectic System.	5	CO-2
(d)	Classify the Polymers on the basis of Tacticity, polymerisation mechanism & on the basis of origin.	5	CO-2